A haptic paradigm to learn how to drive a non-motorised vehicle manipulated through an articulated mechanism

Pierre Martin, Nicolas Férey, Céline Clavel, Patrick Bourdot

VENISE & CPU teams, CNRS/LIMSI, B.P. 133, 91403 Orsay cedex (France)
firstname.name@limsi.fr

Abstract. Virtual Environments are sometimes used to learn how to manipulate some complex and/or dangerous equipments. In this context, we designed a virtual simulator dedicated for the learning of driving a specific vehicle. In our case, the driving task is performed through an articulated mechanism, and requires a physical involvement of the driver. This paper explains how we designed a new haptic paradigm that aims to provide a realistic sensorimotor stimulation, especially in term of physical involvement.

Keywords: Virtual Reality, Haptic Navigation and Driving, Learning.

1. Introduction

Virtual Reality (VR) technologies provide solutions for simulating realistic situations in a controlled and secure context. We explain in this paper the simulation technics and tools used to design an interactive paradigm for the virtual driving of a specific vehicle. We especially focus on a non-motorised forklift, controlled by an articulated handle. The manipulation of this articulated mechanism allows to push/pull the vehicle, and control its direction (Figure 1).
Some works also focus specifically on VR applications dedicated to motorised forklifts. Safety is one of the application fields. [1] studied the impacts on the forklift occurring on drive-in racking structures. They proposed a general method for calculating forces generated under forklift truck impact. In [2] the prevention of forklift capsize is considered. Based on the fact that drivers' training and proper safety procedures are not sufficient enough to reduce accidents, they developed an intelligent control system, embedded on the forklift, which analyzes onboard sensors data and proposes some corrections. Finally, [3] and [4] described a prototype of a full-immersive simulation of forklift truck operations for safety training.

To provide a realistic sensorimotor stimulation during the driving of a non-motorised vehicle manipulated through an articulated mechanism, we designed a new haptic navigation paradigm. It simulates and to haptically renders the articulated mechanism of the handle used to manipulate forklift velocity and direction providing feedback of the ballistic properties through the virtual handle, such as inertia.



Figure 6. The real and virtual non-motorised forklift and articulated handle

2. A haptic virtual mechanism simulating an articulated handle

As explained in the last section, we focus on the virtual driving of a non-motorised vehicle manipulated through an articulated mechanism. Thus, to simulate the real forklift's handle, we designed a virtual mechanism based on a rigid body with hinge constraints (Figure 2 - right). We used a software setup based on *Virtools*™ (a VR platform from Dassault Système) and a specific plugin, called *IPP*™ and developed by Haption, and a hardware setup composed by a *Virtuose*™ *6D* haptic device, also provided by Haption (http://www.haption.com). A first hinge (blue sphere, red arrow) controls the orientation of the forklift's wheels. A second hinge (green arrow), allows user actions to raise or lower the forklift's handle.

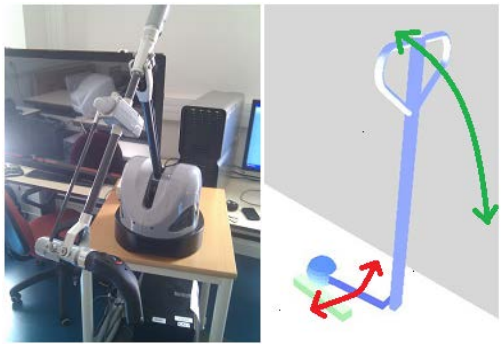


Figure 7. The Virtuose™ 6D 35-45 haptic arm from Haption (left). The virtual mechanism of the forklift handle (right) implemented with the *Interactive Physical Pack* (IPP™/ IPSI™) developed by *Haption* for the *Virtools* platform used for the visual immersion.

3. The Simulation of the ballistic behaviour of the vehicle

This section describes how we compute the ballistic behaviour of the virtual forklift. In the real context, the user walks behind the forklift and controls its velocity by

applying pulling/pushing force through the handle. Then, the velocity of the forklift has to be simulated according to this force. We also have to take into account the weight (400 kg), as well as friction/damping in the real condition, obtained empirically by velocities and deceleration measurements between two points (stopping distance 4.25 m, initial velocity 1 m/s). We express the new velocity of the vehicle according to these factors. Equation 1 is the expression of the new velocity according to F, while Δt is the time variation, t the current time, $V(t)$ the velocity at t, β the damping factor, m the object weight, and α a control constant.

$$V(t+\Delta t) = \left(V(t) e^{\frac{-\beta}{m}\Delta t} + \alpha \frac{F \Delta t}{m} \right) \quad (1)$$

All parameters of Equation 1 are known, except β: this variable takes into account all the forces applied to the object, including the friction forces. In our physical simulation of the forklift, we assume that β is constant, its variation being negligible. To determine its value, we ran the simulation with the forklift moving at a speed of 1m/s, and adapted the value of that constant to make this virtual vehicle stopping at the right distance (the one measured on the real forklift) from its starting point. The only parameter that one must define is α, a constant factor that we could define arbitrarily, in order to be able to scale the force applied on the simulation.

Moreover, we determined the direction of the forklift according to the angle between the green bar and the blue horizontal one (Figure 2 - right). We empirically found the linear mapping between this angle and the inverse of the curvature radius. Using this curvature radius, the current velocity and Equation 1, we were able to compute the next position and orientation of the virtual forklift.

4. Combining a virtual mechanism and a neutral referential technique to control the forklift velocity

The previous sections explain how we haptically render the mechanical behaviour of the forklift, how the direction of the forklift is controlled using the handle angle, and finally how we compute the next position and orientation of the virtual forklift according to the current forklift velocity using Equation 1. We explain in this section how were computed the force used in Equation 1, to control the velocity of the virtual forklift, and which force feedback is provided to the user to simulate a physical involvement during the driving task.

We were inspired by a non-haptic navigation technique based on a non-isomorphic rate-control using the comparisons of a current tracked referential with an initial neutral referential [5]. This technique is especially well adapted to perform navigation task using haptic device, because the base of the haptic device is fixed, and the neutral referential could be chosen as the centre of the haptic workspace.

This concept was thus haptically extended. An elastic force allows the end of the virtual handle described in the previous section (the blue sphere in Figure 2, right) to retrieve the neutral referential placed on the centre of the green bar. The blue sphere is also constrained along this green bar. The opposite of this elastic force is scaled and

used to control the forklift velocity according the Equation 1. This elastic force is also scaled and used as the force feedback provided by the user. It is indeed not possible to provide the same intensity of force feedback through the haptic device (device limitation) than in the real context. The elastic force feedback can also be dynamically linearly scaled according to current velocity, in order to perceive the inertia of the forklift. In this way, more the blue sphere goes forward on the green bar more the force applied by the user on the vehicle is large. The elastic force feedback, depending on the force applied to the virtual forklift and its current velocity, allows our haptic driving paradigm to simulate a physical involvement of the user.

5. Conclusion

We designed an innovative driving paradigm dedicated for the learning of driving a specific vehicle requiring the manipulation of an articulated mechanism and a physical involvement of drivers. On the one hand, the rigid body model provided by *IPP*™ (Haption) into *Virtools*™, allows the users to haptically perceive the physical and mechanical behaviour of the forklift and its handle used to control the velocity and the direction. On the other hand, the velocity control of the virtual forklift is simulated with a push/pull interactive technique using elastic feedback to feel the forklift inertia and to simulate a physical involvement of the user. According to a previous ergonomic study we lead [6], our haptic technique clearly appeared more realistic than the joystick one, considering various dimensions of realism, especially in term of performance and in term of behaviour and psychological processes transfers.

References

[1] Gilbert, B. P., Rasmussen, K. J. R.; (2011) Determination of accidental forklift truck impact forces on drive-in steel rack structures. In Engineer. Struct., 33(5), pp. 1403-1409.

[2] Rinchi, M., Pugi, L., Bartolini, and F. Gozzi, L.; (2010) Design of control system to prevent forklift capsize. Int. J. of Vehicle Systems Modelling and Testing, 5(1), pp. 35–58.

[3] Yuen, K.K., Choi, S.H., and Yang, X.B.; (2010) A full-immersive CAVE-based VR simulation system of forklift truck operations for safety training. In Computer-Aided Design and Applications, 7(2), pp. 235-245.

[4] Bergamasco, M., Perotti, S., Avizzano, C.A., Angerilli, M., Carrozzino, M., Facenza, G., and Frisoli, A.; (2005) Forklift truck simulator for training in industrial environments, Research in Interactive Design: Proc. of Virtual Concept.

[5] Bourdot, P. and Touraine, D.; (2002) Polyvalent Display Framework to Control Virtual Navigations by 6DOF Tracking. In Proc. of IEEE Virtual Reality, pp. 277-278.

[6] Martin, P., Férey, N. Clavel, C., Darses, F. and Bourdot, P.; (2012) Sensorimotor Feedback for Interactive Realism: Evaluation of a Haptic Driving Paradigm for a Forklift Simulator, Lecture Notes in Computer Science, 7282/2012, pp. 314-325.